9. Coupling

The lesson that explains why a Tesla coil is not a transformer

Two tuned circuits handing energy back and forth

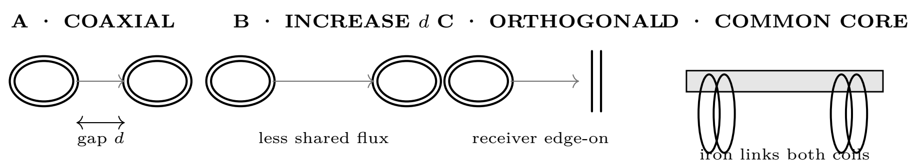
Lesson 9 · about 75 minutes · everyone builds; battery power only

What you need

- Two identical pendulums — same length, same weight
- A horizontal string or dowel to hang them from
- Two identical LC tanks from Lesson 8
- Signal generator and oscilloscope, or the multimeter method
- A ruler, to set and measure coil spacing

Predict first

Two identical pendulums hang from the same slack string. You start one swinging and hold the other still, then let go. What happens to each over the next minute? Draw both, one under the other, against time.



Do it — establish the mechanical model

1. Hang two identical pendulums from a common horizontal string with some slack in it. Start one; leave the other at rest.
2. Every five seconds, record both amplitudes. Continue through one complete transfer out and back. Do not merely write "it moved."
3. Tighten the support string and repeat. Then slacken it and repeat. Graph the two amplitudes against time. Which setting changes transfer time?
4. Shorten one pendulum by 10%, restore the middle string tension, and repeat. This separates frequency mismatch from coupling strength.

Pendulum state record

State	0 s	5 s	10 s	15 s	20 s	transfer time
Matched / slack						
Matched / tight						
Detuned / middle						

Then the coils — geometry before resonance

1. Disconnect both capacitors. Pulse the transmitter coil from the two AA cells through a pushbutton. Connect only the receiver coil to the meter. Tape the leads so they cannot masquerade as a geometry effect.
2. At every state, make five equal presses and record the median absolute receiver pulse. Confirm that the break pulse reverses sign. Stop for warmth.
3. Keep both coil axes coincident and repeat at $d = 0, 1, 2, 4, 8$ cm.

4. Return to $d = 2$ cm. Rotate the receiver about its center through $0°, 30°, 60°, 90°$.
5. Repeat the full distance series with a removable iron core linking both coils. Remove it and recover the original baseline before proceeding.

Repeated distance × core matrix

d (cm)	air: median $\lvert V\rvert$	core: median $\lvert V\rvert$	core/air	spread, anomaly, or control note
0				
1				
2				
4				
8				

Repeated orientation matrix at $d = 2$ cm

θ	median $\lvert V\rvert$	fraction of 0°	projected-area prediction	pulse sign
0°		1.00	maximum	
30°				
60°				
90°			near minimum	

Restore the two tuned tanks

1. Reconnect the identical capacitors and set the coils coaxial at $d = 4$ cm. Sweep across the Lesson 8 resonance in equal frequency steps. Record transmitter and receiver amplitudes.
2. Repeat at $d = 2$ cm and $d = 1$ cm without changing sweep range or drive. Look for one peak becoming two; do not assume it.
3. Change only the receiver capacitor. Repeat at $d = 2$ cm. Then restore the capacitor and verify the original curve returns.

Frequency-sweep matrix

State	f_1	f_2	f_3	f_4	f_5	peak(s)
$d = 4$ cm / matched						
$d = 2$ cm / matched						
$d = 1$ cm / matched						
$d = 2$ cm / detuned						

What it means

The pulse trials isolate mutual inductance before resonance complicates the picture. The sweep trials then show what two matched energy stores do with that shared field. Read the distance, angle, and core graphs as evidence about k; read peak position and splitting as evidence about tuning and exchange rate. Neither graph alone supports both claims.

Coupling coefficient Written k, from 0 (no shared field) to 1 (all of it shared). Here, distance, orientation, and core all change k.

Why loose is better here Energy takes many cycles to cross a loosely coupled pair, and during those cycles the secondary's voltage builds and builds. Couple them tightly and the transfer is fast but the secondary never gets the chance to ring up. Tight coupling also puts so much voltage across the secondary's own turns that it arcs internally and destroys itself.

Both must be tuned the same This is what the pendulum experiment proves. Coupling with mismatched frequencies transfers almost nothing.

Energy goes both ways The pendulums traded back and forth, and so do the coils. In a real coil we want the transfer stopped at the moment the secondary is fullest — which is exactly what the spark gap does in Lesson 10, by quenching.

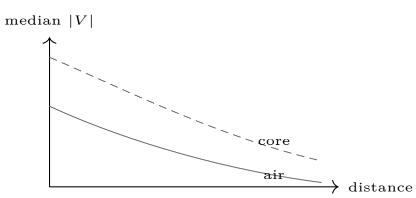
This is why it is not a transformer A transformer trades voltage for current by turns ratio alone. A Tesla coil does that *and* accumulates energy over many cycles of a shared resonance. That second mechanism is where most of the voltage comes from.

Questions that make the graphs earn their ink

1. Which trial is the baseline?
2. What changed when d doubled?
3. Is the distance relation linear?
4. Why keep the axes coincident?
5. Was 90° exactly zero?
6. What stray paths explain a nonzero minimum?
7. Did the core give one constant multiplier?
8. Why compare pulse magnitude, not held voltage?
9. Why does break polarity reverse?
10. Which changes k, and which changes tuning?
11. When did one resonance become two?
12. Did detuning resemble the short pendulum?

Graph, diagnose, prove

Three required graphs. Plot air and core pulse magnitude against distance on shared axes. Plot normalized pulse against angle. Plot receiver amplitude against frequency for every matched spacing and the detuned state. Show measured points; connect them only as guides. Mark readings below resolution and label both split peaks if they exist.



Fault isolation

Symptom	Discriminating check
No pulse	Verify range, receiver continuity, then transmitter current.
Only lead motion works	Tape leads; press without touching coils.
Trials scatter	One operator; inspect button; use the median.
90° stays large	Remove metal; shrink lead loops; recenter.
No resonant peak	Verify each tank alone before coupling them.
Core seems weaker	Restore air baseline; ensure core links both coils.

BLIND FINAL PROOF

A partner hides two receiver states behind a sheet, choosing from coaxial/near, coaxial/far, orthogonal, common-core, or detuned. Take five pulse readings and, where needed, a short frequency sweep. Identify both states and justify them from all three graphs. Swap roles. The proof passes only when both partners identify both states, recover the starting baseline within its measured spread, and observe opposite make/break pulse signs. Diagnose before repeating; never erase a failed trial.

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